



THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

COMP5434 Final Project

Review Ratings Prediction

Group 9

Team Members:

MA Jiyuan (25116696g)

Yee Lok CHIU (25012923g)

YANG Ziting (25050315g)

ZHANG Boxin (25054717g)

Problem definition

Objectives

- Build a scalable distributed data processing pipeline.
- Explore deep learning methods.
- Achieve competitive performance with limited computing resources.

Evaluation Metric

- Root Mean Squared Error (RMSE).
- Formula: $RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$
- Lower RMSE indicates better prediction performance.

Problem definition

Data Analysis

Solution and implementation details

Performance Evaluation and Discussion

Summary of Discoveries and Future Work

Background

- Predict product evaluation scores from 1 to 5 stars.
- Task: Fine-grained regression problem.

Dataset

- Nearly 3 million training records and over 200,000 products.

Data Analysis

Dataset & Data Quality

- **Scale & Framework:** Traditional Pandas is inefficient for millions of records; Apache Spark is required for distributed ETL.
- **Data Quality:** Raw texts contain HTML/URLs. Handled by filling missing texts with "unknown" and numbers with 0.
- **Feature Signals:** Text features show much higher predictive ability than metadata. User-product interactions form a sparse graph.

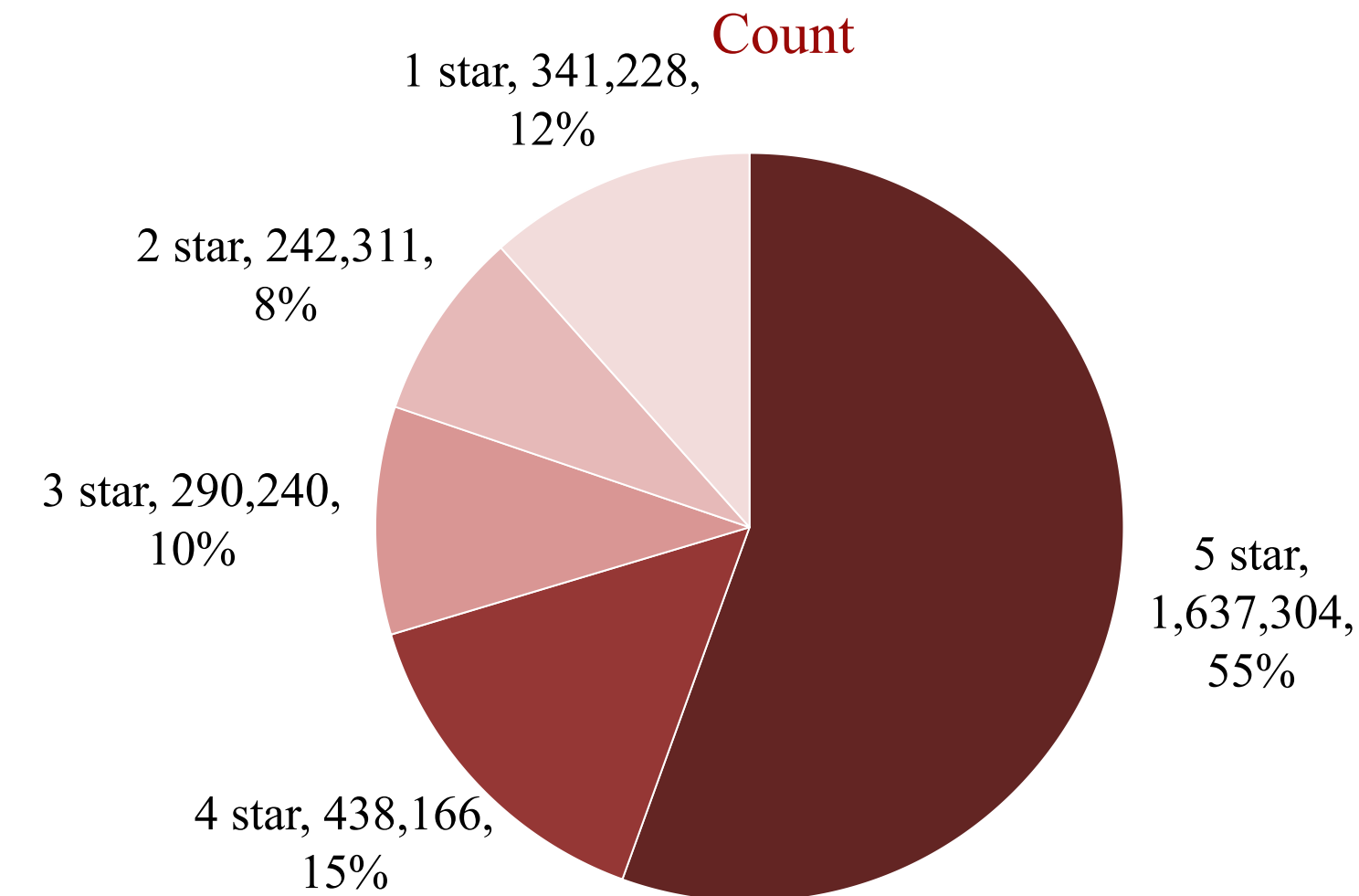


Figure 1.Rating Percentage

Imbalance: 55.52% of reviews give exactly 5 stars.

Impact: Causes distribution compression and makes models favor the majority class.

Problem definition

Data Analysis

Solution and
implementation
details

Performance
Evaluation and
Discussion

Summary of
Discoveries and
Future Work

Overall System Architecture

Data Processing and Model Training Pipeline Architecture

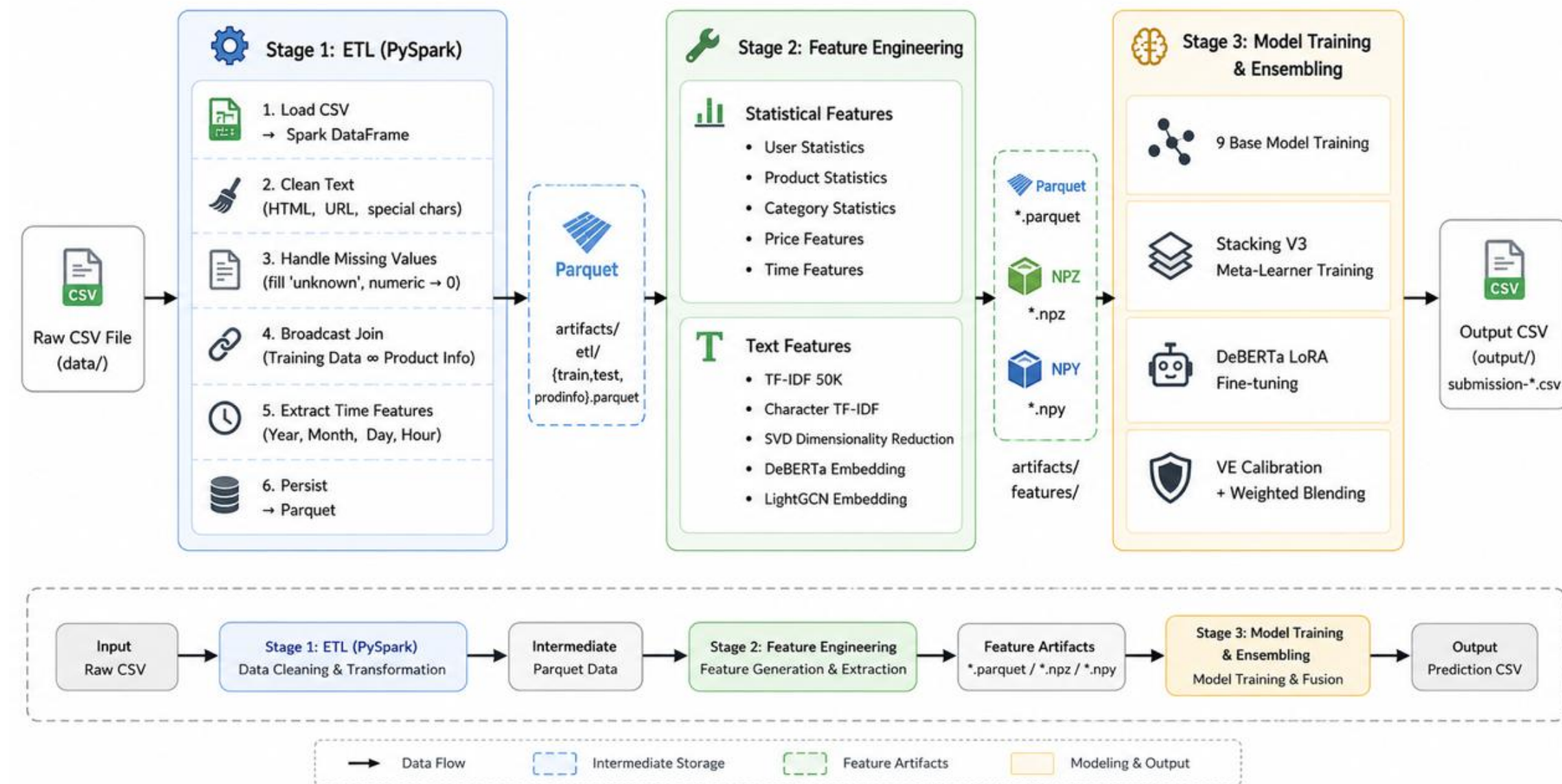


Figure 2. ETL Pipeline

- 3-stage architecture: ETL, Feature Engineering, Model Training.
- Used Apache Spark 3.4.1 for fast distributed processing.
- Used Broadcast Join and Parquet storage to save time and memory.

Problem definition

Data Analysis

Solution and
implementation
details

Performance Evaluation
and Discussion

Summary of
Discoveries and
Future Work

Feature Engineering

Problem definition

Data Analysis

Solution and
implementation
details

Performance Evaluation
and Discussion

Summary of
Discoveries and
Future Work

Statistical

- Capture behavior patterns of users and products.
- Out-of-fold coding strategy strictly prevents target leakage.

Text

- TF-IDF provides an effective sparse lexical baseline.
- SVD dimensionality reduction creates dense feature representations.

Semantic

- DeBERTa embeddings extract deeper semantic context from reviews.

Graph

- LightGCN embeddings model the complex interaction modes between users and products.

DeBERTa Fine-Tuning & Variance Expansion (VE)

Model Design & Fine-Tuning

- DeBERTa-v3-base chosen for strong language understanding.
- Low-Rank Adaptation (LoRA) used to fine-tune parameters efficiently.
- Only 0.32% of the model parameters were updated, saving memory.

Variance Expansion (VE)

The Discovery

- Predicted standard deviation (0.825) < Actual label distribution (1.422).
- Predictions are too focused on the average score.

The Solution

- Variance Expansion (VE): A calibration method to readjust predictions.
- Result: Upgraded Kaggle RMSE from 0.638 to 0.617.

Problem definition

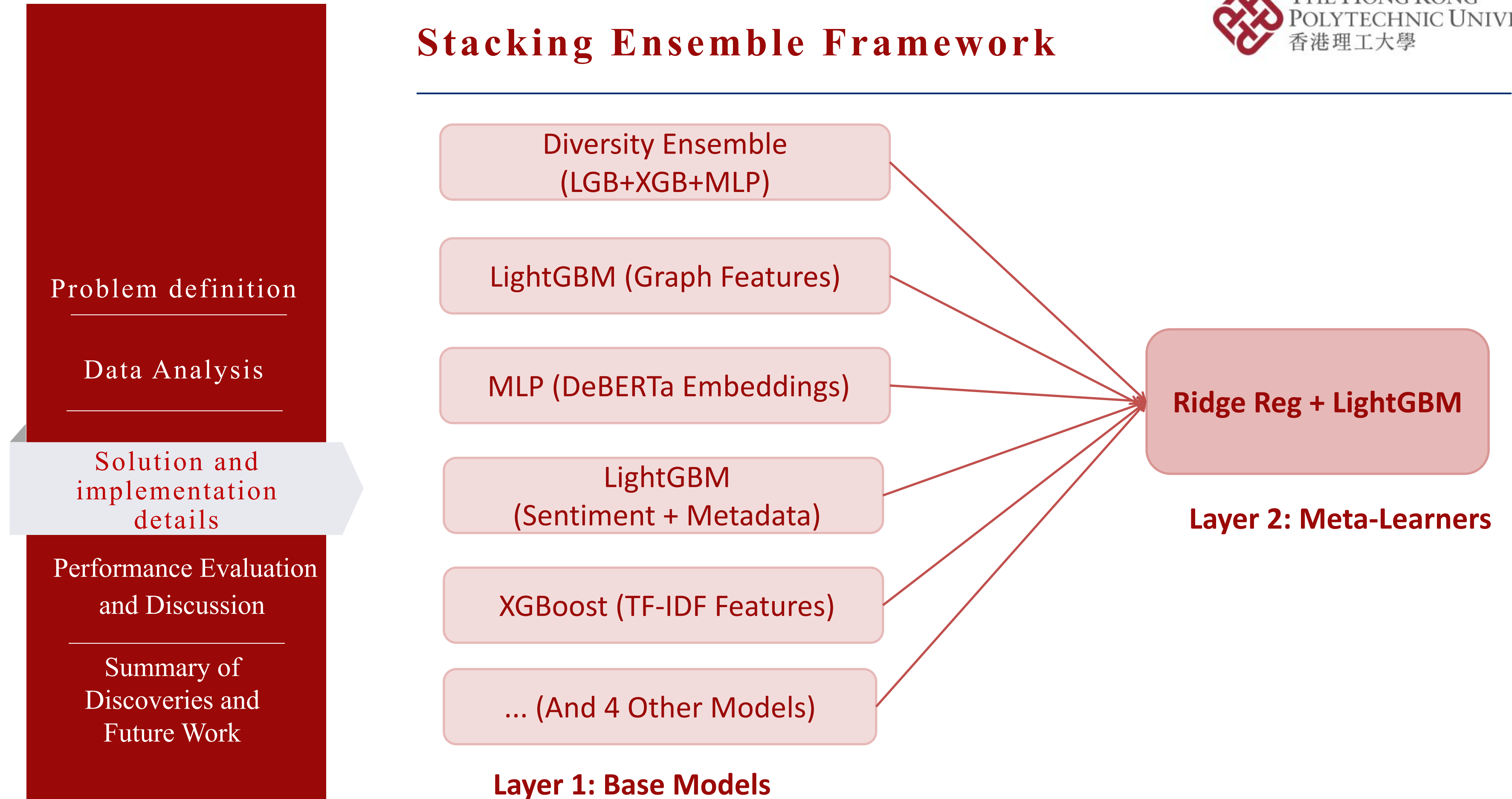
Data Analysis

Solution and
implementation
details

Performance Evaluation
and Discussion

Summary of
Discoveries and
Future Work

Stacking Ensemble Framework



Performance Evaluation

Problem definition

Data Analysis

Solution and
implementation
details

Performance
Evaluation and
Discussion

Summary of
Discoveries and
Future Work

Stage	Method	Kaggle RMSE
Baseline	TF-IDF + LightGBM	0.801
Diversity Ensemble	LGB + XGB + MLP	0.699
Ridge Stacking	6-Model Stacking	0.664
DeBERTa LoRA	Transformer Fine-Tuning	0.638
Variance Expansion	Prediction Calibration	0.617
Final Solution	VE + Stacking V3	0.59770

Table 1. Performance improvement throughout the project

- Final Kaggle RMSE: 0.59770.
- Achieved a 25.4% reduction in RMSE compared with the initial baseline.

- **Optimal Weight Ratio:** 60% VE + 40% Stacking Framework.
- **Hardware:** POLYU HPC 6-Cores CPU, 32GB RAM, 1x NVIDIA RTX 3080 Ti.
- **Key Optimizations:** LoRA fine-tuning & Apache Spark enabled us to process millions of records locally.

Performance Evaluation

Problem definition

Data Analysis

Solution and
implementation
details

Performance
Evaluation and
Discussion

Summary of
Discoveries and
Future Work

Stage	Method	Kaggle RMSE
Baseline	TF-IDF + LightGBM	0.801
Final Solution	VE + Stacking V3	0.59770
Optimal	VE + Stacking V3 + keyword (10 pair)	0.58296

Table 2. Performance improvement by keyword pair

- **Observation:** Final solutions reach a precision bottleneck and are difficult to improve further.
- **Direct Relationship Pairs:** Map explicit review title phrases directly to rating labels: e.g., "Five stars" $\Rightarrow$ 5
 $\Rightarrow$ 4.723(overfit)

- Final Kaggle RMSE: 0.58296.
- Achieved a 27.2% reduction in RMSE compared with the initial baseline.

Major Discoveries

Problem definition

Data Analysis

Solution and
implementation
details

Performance
Evaluation and
Discussion

Summary of
Discoveries and
Future Work

Major Discoveries

- Preventing target leakage is more important than complex features.
- Model set diversity contributes hugely to marginal improvements.
- Predictive calibration (VE) brings massive benefits to imbalanced data.

Future Work

- Train larger transformer models.
- Explore pseudo-labeling with calibration predictions.
- Develop advanced calibration methods and richer graph neural networks.